The Journal of Open Source Software

DeepRiver: A Deep Learning Library for Data Streams

Cedric Kulbach [1¶], **Lucas Cazzonelli** [1], **Hoang-Anh Ngo** [2], **Max Halford** [2], **and Saulo Martiello Mastelini** [3]

1 FZI Research Center for Information Technology, Karlsruhe, Germany **2** AI Institute, University of Waikato, Hamilton, New Zealand **3** University of Sao Paulo, Brazil ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Taher Chegini

Reviewers:

- @musabgultekin
- @atanikan

Submitted: 04 August 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Machine learning algorithms enhance and expedite decision-making processes based on available data. However, as data evolves over time, it becomes crucial to adapt machine learning (ML) systems incrementally to accommodate new data patterns. This adaptation is achieved through online learning or continuous ML technologies. Although deep learning technologies have demonstrated outstanding performance on predefined datasets, their application to online, streaming, and continuous learning scenarios has been limited.

`DeepRiver` is a Python package for deep learning on data streams. Built on top of `River` and `PyTorch`(Paszke et al., 2017), it offers a unified API for both supervised and unsupervised learning. Additionally, it provides a suite of tools for preprocessing data streams and evaluating deep learning models.

Statement of need

In today's rapidly evolving landscape, machine learning (ML) algorithms play a pivotal role in shaping decision-making processes based on available data. The acceleration facilitated by these algorithms necessitates constant adaptation to evolving data structures, as what is valid at one moment may no longer hold in the future. To address this imperative, adopting online learning and continuous ML technologies becomes paramount. While deep learning technologies have demonstrated exceptional performance on static, predefined datasets, their application to dynamic and continuously evolving data streams remains underexplored. The absence of widespread integration of deep learning into online, streaming, and continuous learning scenarios hampers the full potential of these advanced algorithms in real-time decision-making (Kulbach et al., 2024). The emergence of the `DeepRiver` Python package fills a critical void in the field of deep learning on data streams. Leveraging the capabilities of `River`(Montiel et al., 2021) and `PyTorch`(Paszke et al., 2017), `DeepRiver` offers a unified API for both supervised and unsupervised learning, providing a seamless bridge between cutting-edge deep learning techniques and the challenges posed by dynamic data streams. Moreover, the package equips practitioners with essential tools for data stream preprocessing and the evaluation of deep learning models in dynamic, real-time environments. This was already made use of in the context of Streaming Anomaly Detection (Cazzonelli & Kulbach, 2022). As the demand for effective and efficient adaptation of machine learning systems to evolving data structures continues to grow, the integration of `DeepRiver` into the landscape becomes crucial. This package stands as a valuable asset, unlocking the potential for deep learning technologies to excel in online, streaming, and continuous learning scenarios. The need for such advancements is evident in the quest to harness the full power of machine learning in dynamically changing environments, ensuring our decision-making processes remain accurate, relevant, and agile in the face of evolving data landscapes.

Features

DeepRiver enables the usage of deep learning models for data streams. This means that deep learning models need to adapt to changes within the evolving data stream [Bayram et al. (2022)](Lu et al., 2018) i.ex. the number of classes might change over time. In addition to the integration of PyTorch(Paszke et al., 2017) into River(Montiel et al., 2021), this package offers additional data stream specific functionalities such as class incremental learning or specific optimizers for data streams.

Compatibility

DeepRiver is built on the unified application programming interface (API) of River(Montiel et al., 2021) that seamlessly integrates both supervised and unsupervised learning techniques. Further, it integrates the huge functionality of PyTorch(Paszke et al., 2017) for deep learning such as using GPU acceleration and a broad range of architectures. This unified approach simplifies the development process and facilitates a cohesive workflow for practitioners working with dynamic data streams. Leveraging the capabilities of the well-established River(Montiel et al., 2021) library and the powerful PyTorch(Paszke et al., 2017) framework, DeepRiver combines the strengths of these technologies to deliver a robust and flexible platform for deep learning on data streams. This foundation ensures reliability, scalability, and compatibility with state-of-the-art machine learning methodologies. It provides comprehensive documentation to guide users through the installation, implementation, and customization processes. Additionally, a supportive community ensures that users have access to resources, discussions, and assistance, fostering a collaborative environment for continuous improvement and knowledge sharing.

Adaptivity

DeepRiver is specifically designed to cater to the requirements of online learning scenarios. It enables continuous adaptation to evolving data by supporting incremental updates and learning from new observations in real time, a critical feature for applications where data arrives sequentially. Further, it enables the model to adapt to changes in the number of classes over time for classification tasks. It equips practitioners with tools for evaluating the performance of deep learning models on data streams. This feature is crucial for ensuring the reliability and effectiveness of models in real-time applications, enabling users to monitor and fine-tune their models as the data evolves.

Architecture

The DeepRiver library is structured around various types of estimators for anomaly detection, classification, and regression. Each category contains several specialized classes that inherit from more general base classes, forming a hierarchical structure. In anomaly detection, the base class AnomalyScaler has derived classes AnomalyMeanScaler, AnomalyMinMaxScaler, and AnomalyStandardScaler. Additionally, the Autoencoder class, which inherits from DeepEstimator, has a specialized subclass called ProbabilityWeightedAutoencoder. The RollingAutoencoder class inherits from RollingDeepEstimator.

For classification, the base class Classifier inherits from DeepEstimator. Derived from Classifier are specific classes like LogisticRegression and MultiLayerPerceptron. The RollingClassifier class inherits from both RollingDeepEstimator and Classifier.

In regression, the base class Regressor inherits from DeepEstimator. Specific regression classes like LinearRegression and MultiLayerPerceptron inherit from Regressor. The MultiTargetRegressor also inherits from DeepEstimator. The RollingRegressor class inherits from both RollingDeepEstimator and Regressor.

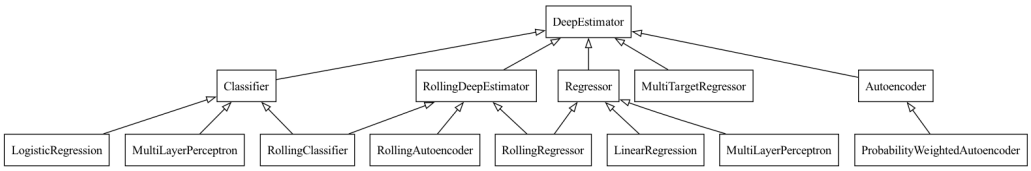

Figure 1: Architecture of DeepRiver

Overall, the library is organized to provide a flexible and hierarchical framework for different types of machine learning tasks, with a clear inheritance structure connecting more specific implementations to their base classes.

Acknowledgements

This work was carried out with the support of the Research Center for Information Technology in Karlsruhe, Germany.

References

Bayram, F., Ahmed, B. S., & Kassler, A. (2022). From concept drift to model degradation: An overview on performance-aware drift detectors. *Knowledge-Based Systems*, *245*, 108632. https://doi.org/10.1016/j.knosys.2022.108632

Cazzonelli, L., & Kulbach, C. (2022). Detecting anomalies with autoencoders on data streams. *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, 258–274. https://doi.org/10.1007/978-3-031-26387-3_16

Kulbach, C., Cazzonelli, L., Ngo, H.-A., Le-Nguyen, M.-H., & Bifet, A. (2024). *A retrospective of the tutorial on opportunities and challenges of online deep learning*. https://doi.org/10.48550/arXiv.2405.17222

Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., & Zhang, G. (2018). Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, *31*(12), 2346–2363. https://doi.org/10.1109/TKDE.2018.2876857

Montiel, J., Halford, M., Mastelini, S. M., Bolmier, G., Sourty, R., Vaysse, R., Zouitine, A., Gomes, H. M., Read, J., Abdessalem, T., & Bifet, A. (2021). River: Machine learning for streaming data in python. *Journal of Machine Learning Research*, *22*(110), 1–8. http://jmlr.org/papers/v22/20-1380.html

Paszke, A., Gross, S., Chintala, S., Chanan, G., Yang, E., DeVito, Z., Lin, Z., Desmaison, A., Antiga, L., & Lerer, A. (2017). Automatic differentiation in PyTorch. *NIPS-w*.

DRAFT